

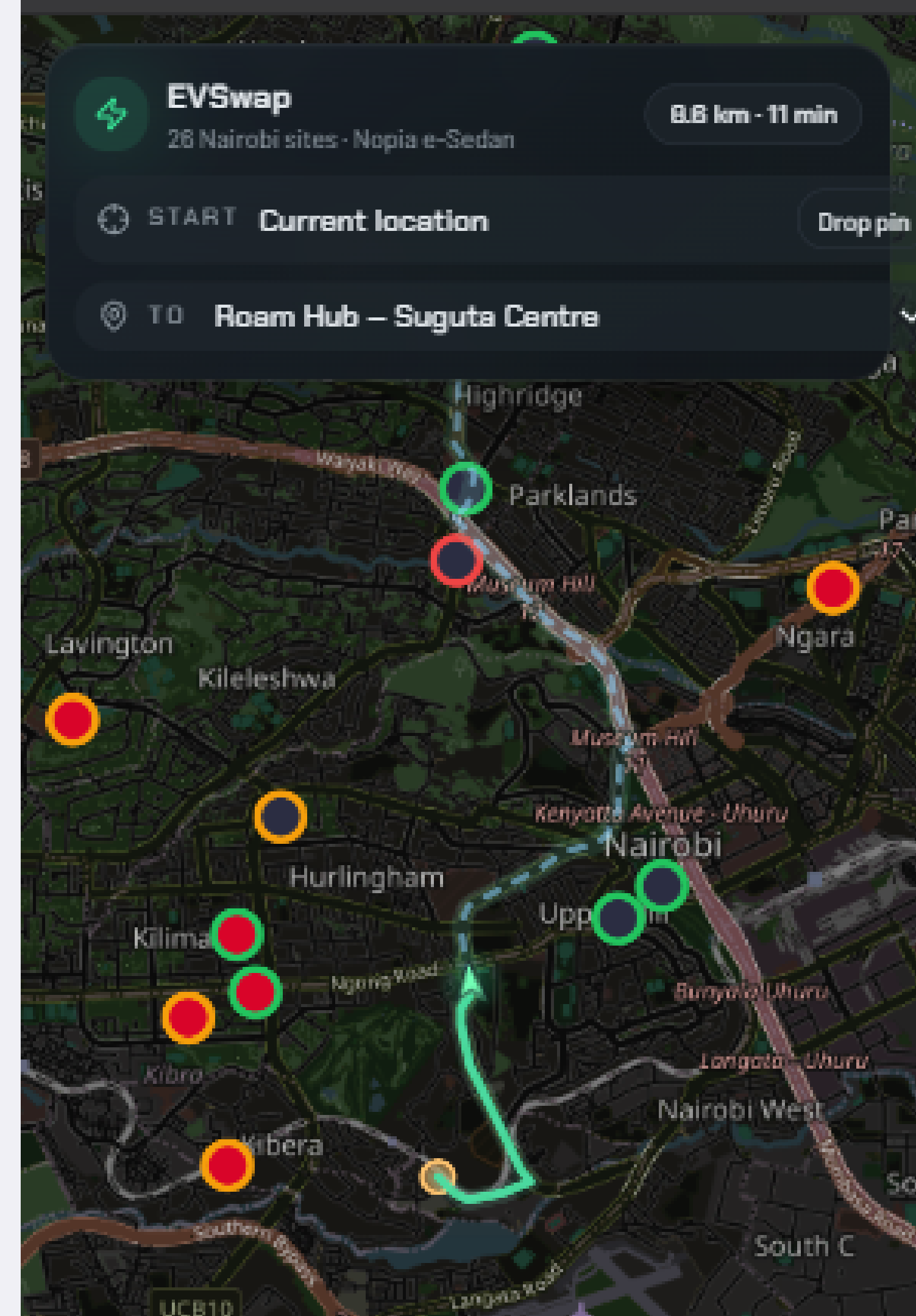
BUILT IN NAIROBI, BUILT FOR THE WORLD

# EVSWAPP

A replicable AI-driven model that leverages city data to accelerate urban e-mobility through intelligent infrastructure siting.

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AI X CITY HACKATHON 2026



## 02 | THE PROBLEM

Nairobi's transport sector is a major, growing emissions source.

- ⚡ E-mobility (buses, e-bikes, boda-bodas) is the fastest mitigation lever available today.
- 📍 Infrastructure is currently sited by **commercial logic**, not transit need.
- ⚠️ Adoption stalls wherever access lags, creating a cycle of exclusion.

TRANSPORT SECTOR  
CONTRIBUTION

**MAJOR**

EMISSIONS SOURCE

## 02 | CONTEXT

# Why Nairobi

*A representative “high-informality, high-growth” African city. The hard case, on purpose. If it works here, it generalizes.*

**4.6M**

residents (WorldPop 2020,  
verified in this project)

**136**

matatu routes -  
headway-based, no fixed  
timetable

**4,284**

matatu stops  
citywide

**865**

H3 hexagons  
tessellating the city

Nairobi's matatu network runs on informal, driver-set headways rather than a published schedule. This project derives frequency directly from real GTFS trip data (Digital Matatus), not assumptions.

## 03 | OUR SOLUTION

# An AI-driven, replicable e-mobility siting model.

We score every part of Nairobi for charging need using real population + transit data, recommending phased sites per the **Avoid-Shift-Improve** framework.



### DATA-LED SITING

Leveraging real-world population and transit demand data to identify high-impact locations.

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### PHASED ROLLOUT

Strategic prioritization: Electrifying buses first, followed by e-bikes and boda-bodas.

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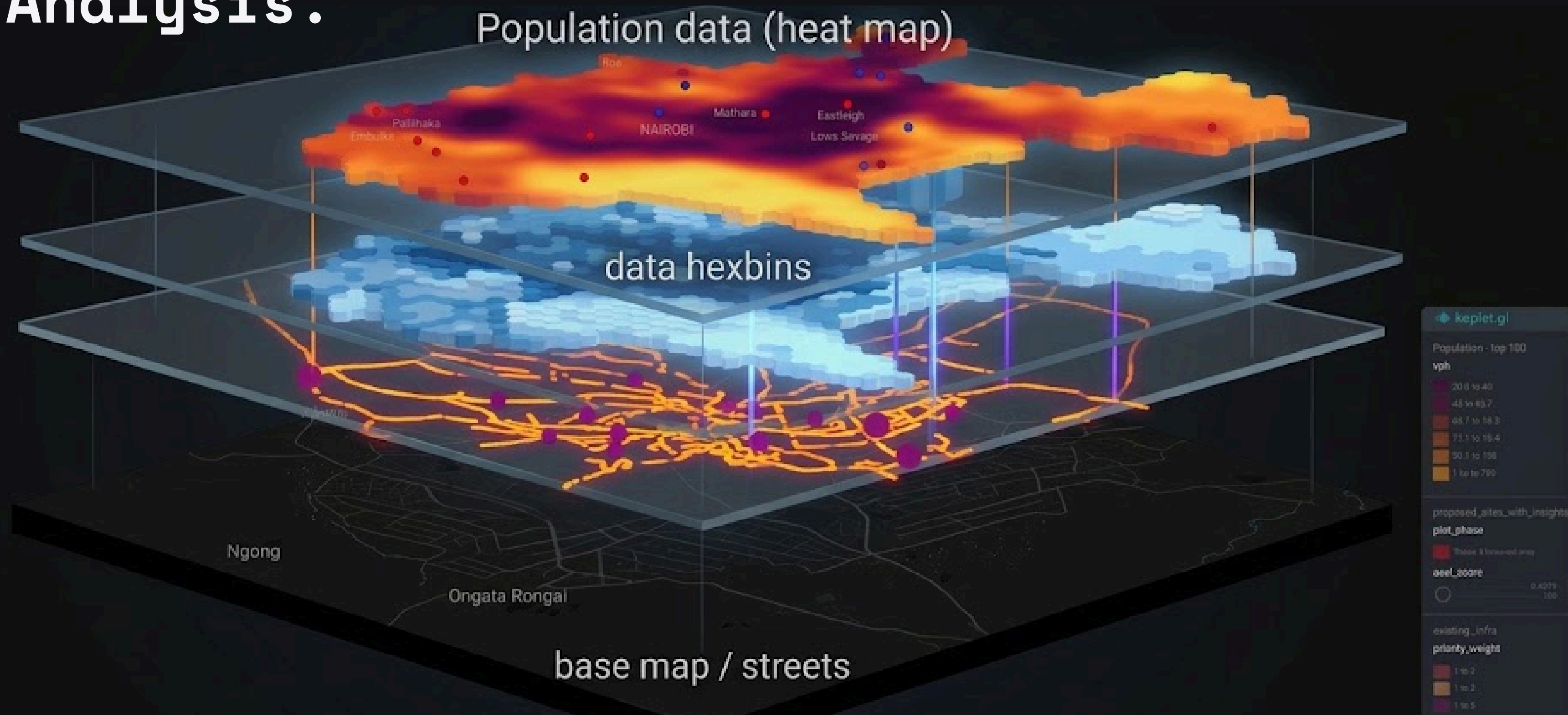
### GLOBAL REPLICABILITY

Open data pipelines that can be deployed in any urbanizing city.

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## 04 | METHODOLOGY

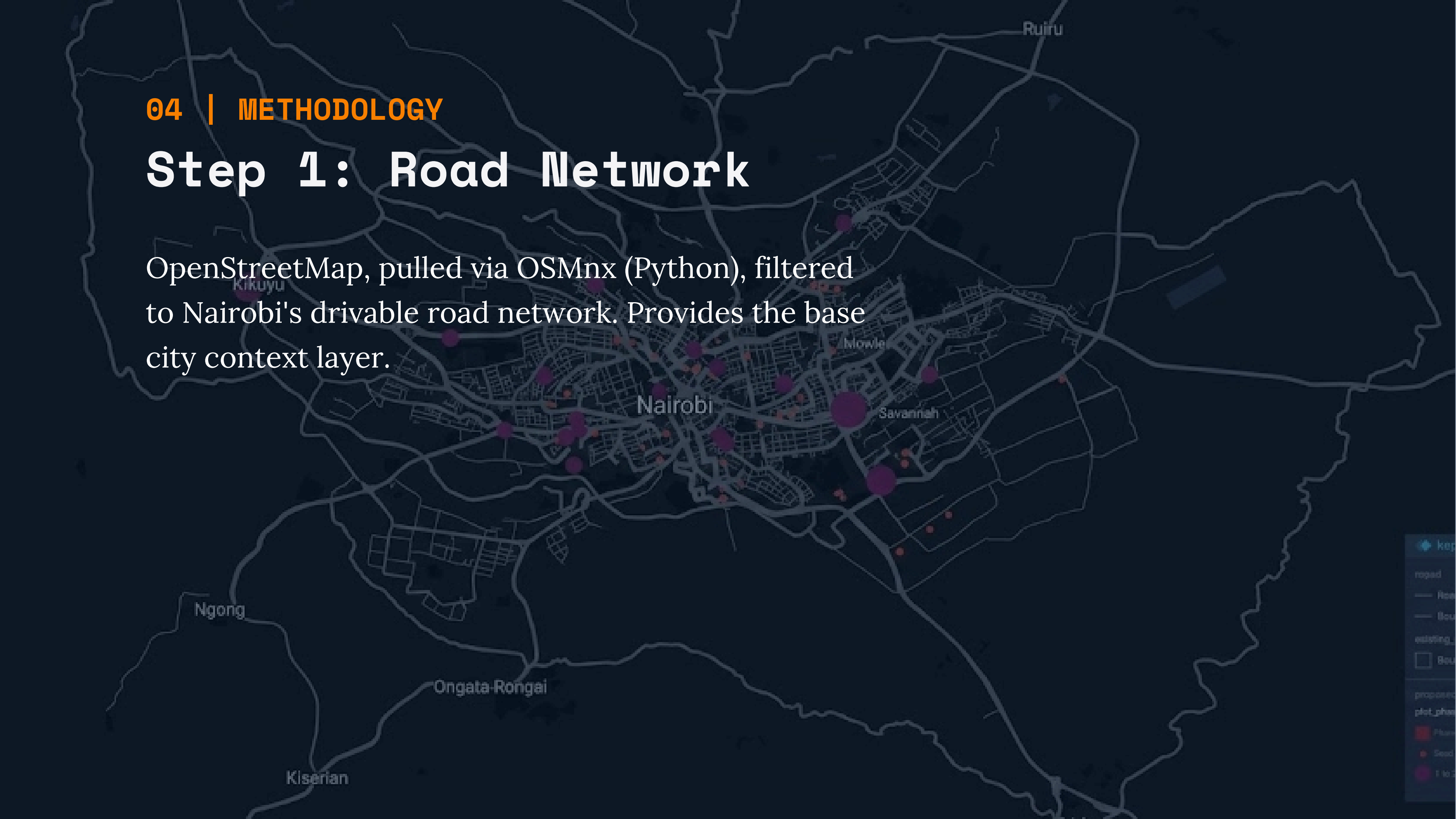
# Four Step Spatial Methodology for City Analysis.



## 04 | METHODOLOGY

### Step 1: Road Network

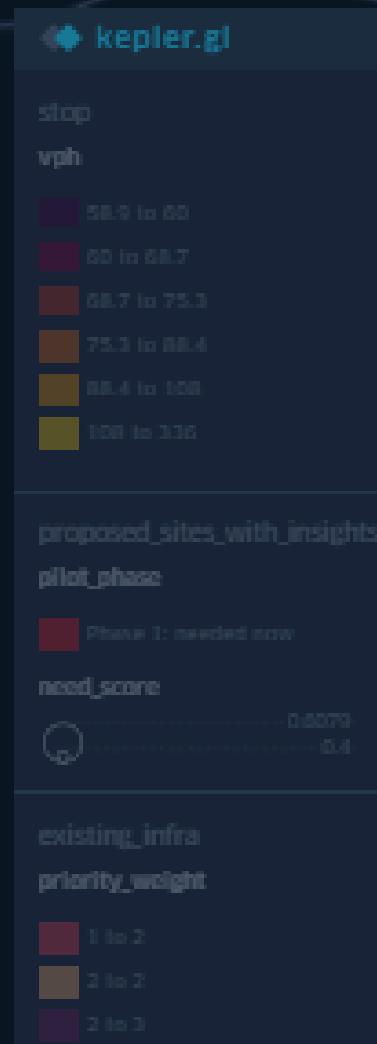
OpenStreetMap, pulled via OSMnx (Python), filtered to Nairobi's drivable road network. Provides the base city context layer.



## 04 | METHODOLOGY

# Step 2: Population Data

WorldPop's 100m-resolution, constrained, UN-adjusted gridded population estimate for Kenya (2020). "Constrained" means it estimates population only where actual building footprints exist. Zonal-summed into each hexagon. (Total came out to 4.62M; consistent with Nairobi's known population.)



## 04 | METHODOLOGY

### Step 3: Transport Patterns

The Digital Matatus GTFS feed for Nairobi (136 routes, 4,284 stops), sourced via TUMI Datahub. Peak/off-peak derived from the feed's own frequencies.txt defines three real time bands (AM peak 06:00–09:00, midday 09:00–15:00, PM peak 15:00–21:00), which we used directly to compute vehicles-per-hour per stop per band.

## 04 | METHODOLOGY

# Step 4: Multi Criteria Need-analysis

A composite model per hexagon: 60% population density + 40% peak transit demand (both min-max normalized), discounted by existing e-mobility charging/swap coverage.



## 05 | THE AI, MADE VISIBLE

# Transparent intelligence, not a black box.

01

### NEED-SCORING MODEL

Multi-criteria using a transparent, auditable spatial analysis form

```
Score = (Pop +  
Transit_D) -  
Coverage
```

02

### GROWTH FORECASTING

WorldPop CAGR-based phased urgency to stay ahead of urban expansion.

Predictive Horizon:  
2026

03

### LIVE LLM INSIGHTS

Plain-language, data-grounded rationale generated in real time for every site.

*"High transit demand at this node justifies a 10-bay battery swap station..."*

## AI INTEGRATION

# Multi-Criteria Need-Analysis Model

```
demand_score = 0.6 × pop_norm + 0.4 ×  
transit_norm
```

```
existing_coverage =  $\sum$  priority_weight  
(point stations in hex)
```

```
need_score = demand_score × (1 -  
coverage_norm)
```

865

hexagons ranked by  
e-mobility infrastructure need

## 06 | WHY THIS IS REPLICABLE

# A Global Pipeline for World Cities

[Github link](#)



### OPENSTREETMAP (OSM)

Global road network and infrastructure data, accessible for any urban center.



### WORLDPOP

High-resolution contemporary data on human population distributions.



### GTFS FEEDS

Standardized transit schedules and geographic information for city transport.



**City-agnostic scripts: Reproducible with AI coding assistants, no specialist GIS department required.**

**Nairobi's highest-need  
e-mobility hotspots have**

**ZERO**

**verified charging coverage  
today.**

 **CURRENT INFRASTRUCTURE**

Clusters heavily around commercial and CBD sites, following high-income real estate trends rather than transit demand.

 **ACTUAL MOBILITY NEED**

Concentrates in high-density residential corridors and peak transit hubs where charging access is non-existent.

## 08 | THE GAP IS GROWING

Urban growth is outpacing infrastructure.

+21,000

Predicted new residents in the #1 need hexagon by mid-2026.

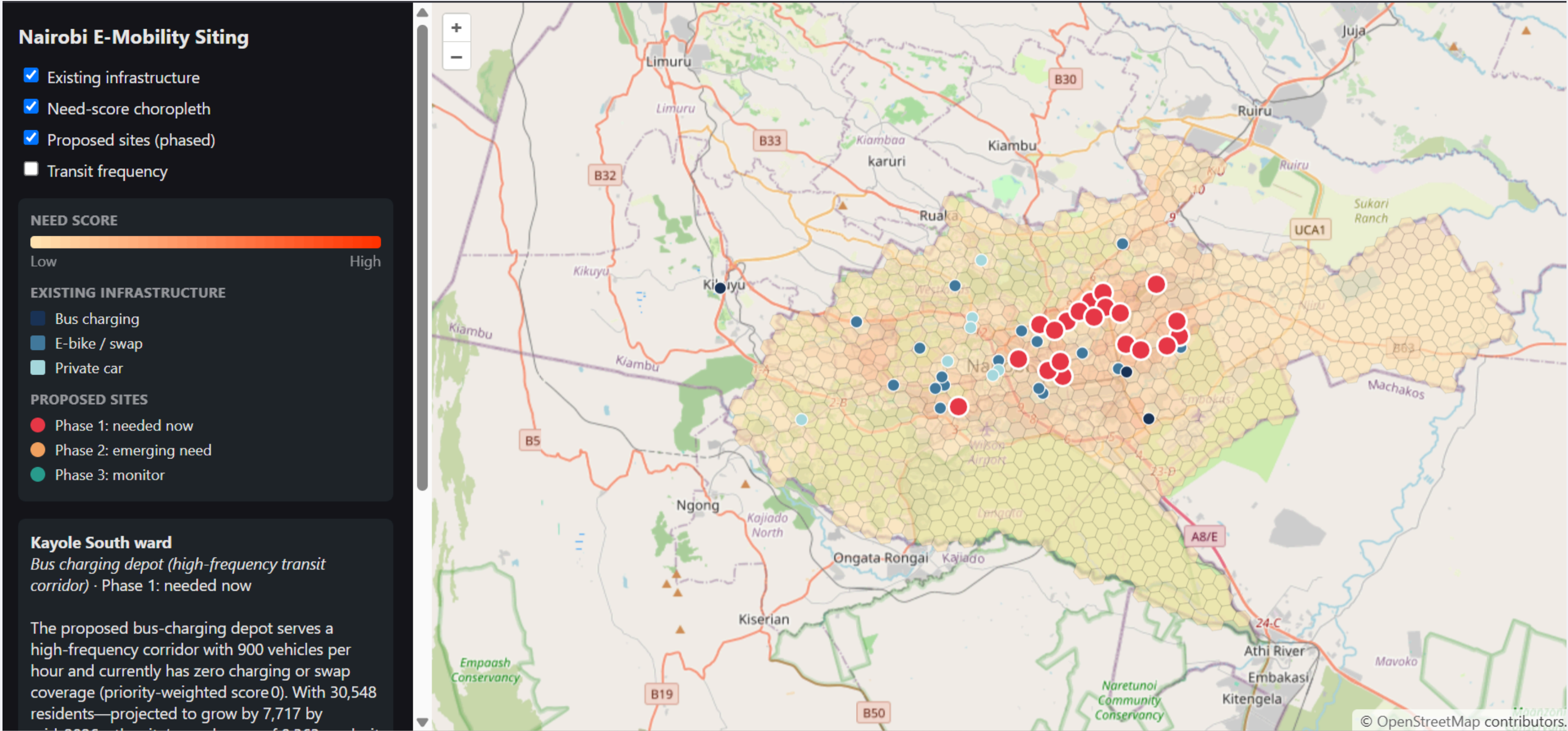
### COMPOUNDING RISK

Current mobility need and fastest population growth overlap perfectly.

Without data-driven intervention, the access gap compounds every month, locking in high-emission transit patterns for new residents.

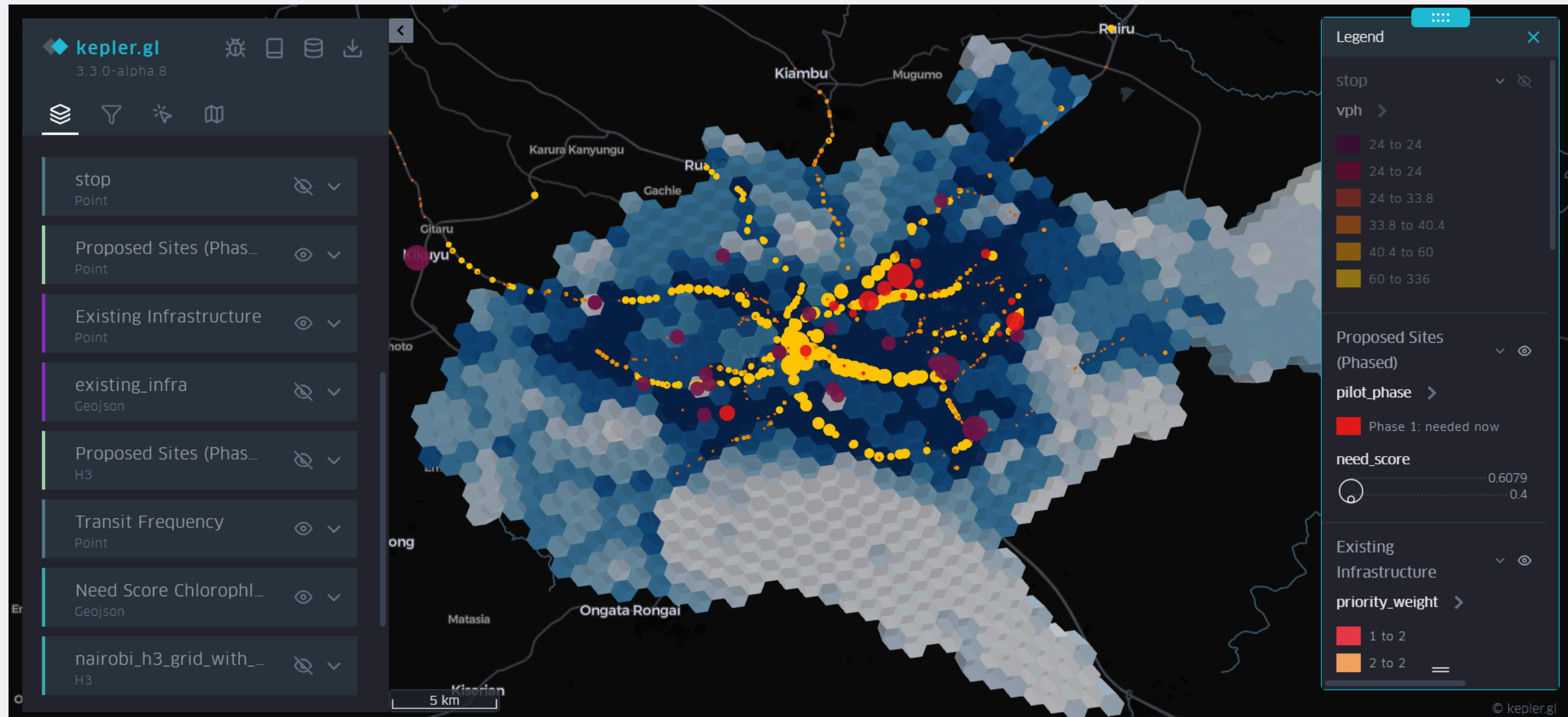
🕒 CRITICAL WINDOW: NEXT 18 MONTHS

# Proposed Sites + Live AI Insights



## 09 | LIVE DEMO PREVIEW

# Fully interactive insights Dashboard.

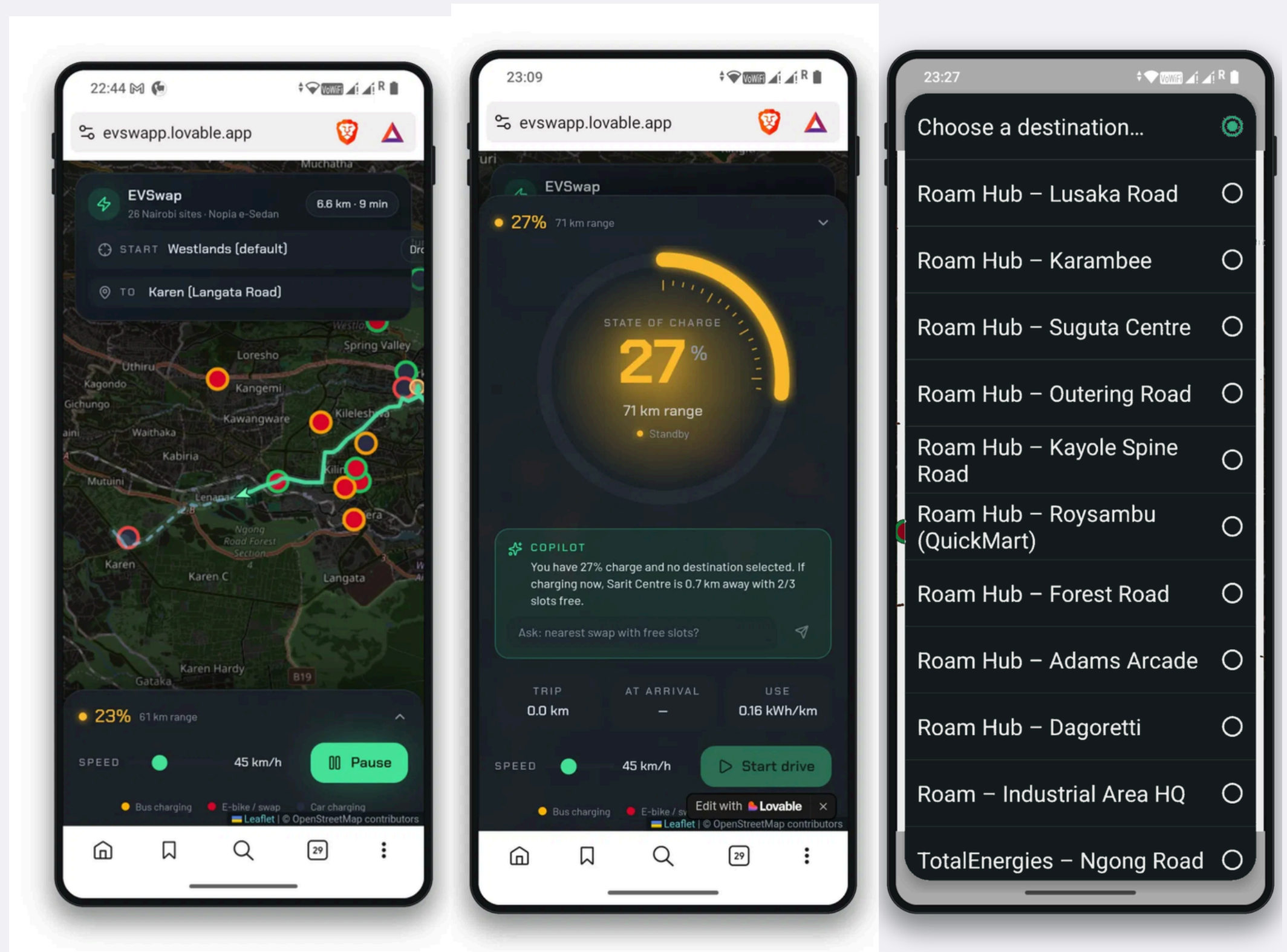


Designed for decisions, not just discovery.

LIVE PROTOTYPE AVAILABLE

09 | LIVE DEMO PREVIEW

# Mobile-First EVSwapp App



Designed for decisions, not just discovery.

[evswapp.lovable.app](https://evswapp.lovable.app)

## 10 | WHO THIS IS FOR

# One decision layer. Multiple buyers.



### CITY PLANNING & TRANSPORT

Data-driven urban policy



### NATIONAL POLICY TEAMS

E-mobility roadmaps



### C40 / I4C MEMBER CITIES

Global replicability



### E-MOBILITY OPERATORS

Roam, BasiGo, Ampersand



### REAL ESTATE OWNERS

Asset monetization



### CLIMATE FINANCE

Impact investors

## 11 | BUSINESS MODEL & IMPACT

# Here is How we Make Money

### SITING-AS-A-SERVICE

Charge-point operators pay for verified, ranked, AI-explained site recommendations, turning high-risk capex into data.

**MODEL: PER-REPORT / SUB**

### REAL ESTATE MATCHMAKING

Property owners list sites; we score them against demand and match them to operators for a conversion fee.

**MODEL: BROKERAGE FEE**

### ADVERTISING INTELLIGENCE

High-need sites are high-traffic sites. We license footfall data and take a share of station digital-screen ad revenue.

**MODEL: DATA LICENSE / REV SHARE**

### TARGET POPULATION

Underserved Phase-1 Hotspots

### MODE SHIFT

Petrol Bodaas → E-Bikes & Buses

### ALIGNMENT

C40 Transport-Emissions Goals

*"Every stream monetizes the same core asset – the need-score data – sold to a different buyer for a different decision."*

## 12 | OUR ASK & THE TEAM

# We Request Support to Develop and Pilot EVSwapp With a Matched City.



**Priscila Adhiambo**

GIS Engineer | Urban Planner



**Fauzia Omala**

Software Engineer



**Benard Kimani**

AI/ML Engineer



**Chibuike Cletus**

Project Manager



**Lynne Adhiambo**

Urban Designer

**Built in Nairobi, Built for the World.**